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Quest 9: The Sulphurprise

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Part I

Story section

While the Ent warriors were growing, we worked on another part of the plan. An army alone would not be enough. We still needed a way to get inside Nulldor and destroy it from within, along with the egg.

The plan was simple, at least in theory. The Ents would attack from the front and keep everyone busy, while our team slipped into Nulldor through the old sewers beneath it. Once inside, we would need something a little more destructive.

Our weapon was a special plant that normally grew very slowly but reacted violently to sulphur. With enough sulphur, its branches grew at an astonishing rate, splitting again and again as they spread.

Fortunately, if there was one thing we expected to find plenty of inside a place like Nulldor, it was sulphur. The only remaining problem was making sure we understood exactly how the plant would grow before carrying it into the enemy stronghold.

We had prepared several varieties of the plant and recorded their properties (your notes). Each line described one variety:

`id=ID, growth=GROWTH, limit=LIMIT`

At first, each plant was nothing more than a seed. As soon as sulphur reached it, a single branch began growing from the seed. **Every existing branch then extended by one element every second**, with all branches growing at the same time.

Each variety had its own critical `growth` value, which was always odd. Whenever a branch reached that size, it split exactly in the middle. The middle element became a branching point, and the original branch became two branches of equal length, with the branching point between them.

At the same moment, a new branch, already one element long, grew from that branching point. From the next second onwards, all three branches continued growing independently, and any of them could eventually split in exactly the same way.

To keep the experiments under control, each plant had its own `limit`. Every element belonging to a branch counted towards the plant's size. A branching point remained part of the plant and also counted as one element, although it was no longer part of any branch. The original seed itself did not count.

The experiment stopped at the end of the first second when the total size of the plant reached or exceeded its limit. I needed to find how many seconds this took for each plant and **add all those times together.**

Example based on the following notes:

```
id=1, growth=3, limit=10
id=2, growth=5, limit=20
id=3, growth=7, limit=30
```

For plant `1`, each branch splits as soon as it reaches `3` elements. Its growth can be represented as follows:

```
                                  \ / \ /
                                   +   +
                         \   /      \ /
                          \ /        +
                  \ /      +         \ /
             |     +       |          +
   .    |    |     |       |          |

   0    1    2     3       4          5
```

The total number of elements after each second is:

```
Second:    1   2   3   4   5
Elements:  1   2   4   7  13
```

The limit of `10` is first reached or exceeded after `5` seconds, so plant `1` grows for `5` seconds.

Applying the same rules to plant `2`, where branches split after reaching `5` elements, gives:

```
                                                                        \
                                                                         \   /
                                                                          \ /   \
                                                                           +     \ /
                                                          \                 \     +
                                                           \ /     /         \   /
                                              \             +     /           \ /
                                               \     /       \   /             +
                                    \           \   /         \ /              |
                                     \   /       \ /           +               | /
                            \         \ /         +            |               |/
                       |     \ /       +          |            |/              +
                  |    |      +        |          |            +               |
             |    |    |      |        |          |            |               |
   .    |    |    |    |      |        |          |            |               |

   0    1    2    3    4      5        6          7            8               9
```

The total number of elements after each second is:

```
Second:    1   2   3   4   5   6   7   8   9
Elements:  1   2   3   4   6   9  12  17  25
```

The limit of `20` is first reached or exceeded after `9` seconds.

For plant `3`, branches split after reaching `7` elements. Its total size develops as follows:

```
Second:    1   2   3   4   5   6   7   8   9  10  11  12  13
Elements:  1   2   3   4   5   6   8  11  14  17  22  29  37
```

The limit of `30` is first reached or exceeded after `13` seconds.

The growth times are therefore:

```
Plant 1:  5 seconds
Plant 2:  9 seconds
Plant 3: 13 seconds
```

The required sum is:

5 + 9 + 13 = **27**

What was the sum of the growth times for all plants?

Your notes for this part:

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Part I solved with answer: 209

Check your progress

Part II

After watching the first experiments, we decided that total plant size was not the best way to keep its growth under control. A long branch could contain many elements without making the plant much harder to handle. What really worried us was the number of directions in which it could keep growing.

So we changed our approach. **Instead of limiting the total number of elements, we limited the number of leaves.** A leaf was the growing tip at the end of a branch. After the first second, the plant had one leaf, and every new branch created during a split added one more.

The growth rules remained exactly the same. Every branch extended by one element per second, and branches split when they reached their `growth` value. This time, however, the `limit` value described the maximum number of leaves.

The experiment stopped at the end of the first second when the number of leaves reached or exceeded the limit.

Using the new plant definitions (your notes), I needed to determine how many seconds each plant grew before reaching its leaf limit, and **add all those times together**.

Example based on the following notes:

```
id=1, growth=3, limit=10
id=2, growth=5, limit=20
id=3, growth=7, limit=30
```

For plant `1`, the shape develops in exactly the same way as before:

```
                                                            \ / \ / \ / \ /
                                                             +   +   +   +
                                                              \ /     \ /
                                                               +       +
                                             \   / \   /        \ / \ /
                                              \ /   \ /          +   +
                                               +     +            \ /
                                                \   /              +
                                  \ / \ /        \ /                \ / \ /
                                   +   +          +                  +   +
                         \   /      \ /            \   /              \ /
                          \ /        +              \ /                +
                  \ /      +         \ /             +                 |/
             |     +       |          +              |                 +
   .    |    |     |       |          |              |                 |

   0    1    2     3       4          5              6                 7
```

This time, only the leaves at the ends of the branches matter. The number of leaves after each second is:

```
Second:  1   2   3   4   5   6   7
Leaves:  1   1   2   2   5   5  14
```

The limit of `10` leaves is first reached or exceeded after `7` seconds, when the plant has `14` leaves.

Applying the same rules to plant `2` gives:

```
Second:  1   2   3   4   5   6   7   8   9  10  11  12  13  14
Leaves:  1   1   1   1   2   2   2   4   5   5   9  13  14  22
```

Its limit of `20` leaves is first reached or exceeded after `14` seconds.

For plant `3`, the leaf count develops as follows:

```
Second:  1   2   3   4   5   6   7   8   9  10  11  12  13  14  15  16  17  18  19  20  21
Leaves:  1   1   1   1   1   1   2   2   2   2   4   4   5   5   9   9  13  13  22  22  34
```

Its limit of `30` leaves is first reached or exceeded after `21` seconds.

The growth times are therefore:

```
Plant 1:  7 seconds
Plant 2: 14 seconds
Plant 3: 21 seconds
```

The required sum is:

7 + 14 + 21 = **42**

What was the sum of the growth times for all plants?

Your notes for this part:

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Part II solved with answer: 15446

Check your progress

Part III

Our new way of controlling the plants seemed to work. Watching the number of leaves gave us a much better idea of when the growth became dangerous. More importantly, none of the experiments had escaped the laboratory yet.

There was only one thing left to check before we trusted the weapon inside Nulldor. The small experiments had behaved exactly as expected, but the real plants would grow on a much larger scale. We prepared another set of varieties (your notes), this time covering a much wider range of `growth` and `limit` values, including some considerably larger than before.

All the rules from Part II remained exactly the same. Every branch grew by one element per second, branches split in the same way, and the `limit` still referred to the number of leaves. The experiment still ended after the first second in which the number of leaves reached or exceeded that limit.

Actually growing plants of this size did not seem like a particularly good idea. Fortunately, our earlier experiments had given us enough information to **calculate their growth efficiently without simulating every single second**.

What was the sum of the growth times for all plants?

Your notes for this part:

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Part III solved with answer: 849559434880880

Check your progress

Puzzle solved! Don't stop now!

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